

Microwave radiometer data quality assurance at Magurele station, Romania

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1. Microwave Radiometer Calibrations

Introduction

The Microwave Radiometer (MWR) is a passive remote sensing device designed to detect radiance emitted naturally by the atmosphere within specified frequency bands ranging from 20 to 60 GHz (Westwater et al., 2004). MWR technology has reached a level of maturity and is widely utilized for extracting atmospheric temperature and humidity profiles (Rose et al., 2005), as well as measurements of integrated water vapor and liquid water content. MWRs offer retrieval capabilities under various sky conditions, including both clear and cloudy skies, with a high temporal resolution. They typically provide moderate vertical resolution, with most of the information concentrated within the Planetary Boundary Layer (PBL). MWR channels positioned near the 60 GHz oxygen complex are utilized for retrieving temperature profiles, while those in proximity to the 22.235 GHz water vapor line offer data on humidity, integrated water vapor, and are also sensitive to the column-integrated liquid water content.

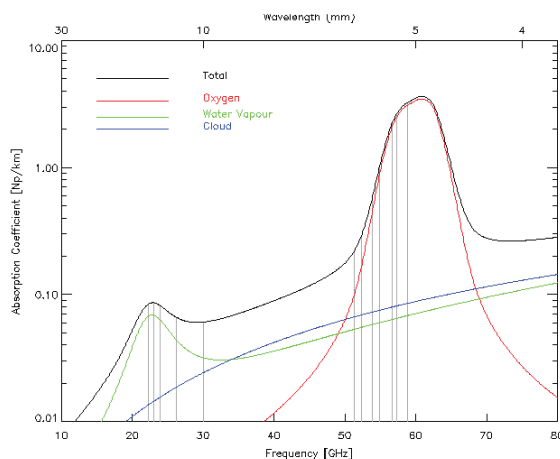
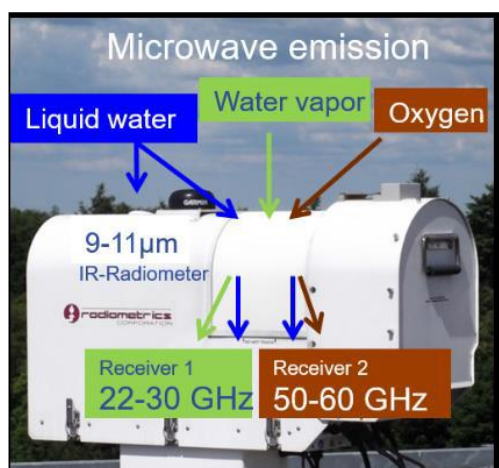


Figure 1. Microwave Radiometer - Constitution of the overall absorption coefficient in a typical atmospheric state (DWD Pilotstation Evaluating MWRs for future observing networks, Christine Knist, Moritz Löffler, Jasmin Vural, Annika Schomburg, Deutscher Wetterdienst, Germany)

Microwave radiometers serve as passive instruments detecting radiances from atmospheric emissions due to gases and hydrometeors, converting these radiances into voltages. Regular calibration of these instruments is imperative to maintain data quality. For HATPRO radiometers manufactured by Radiometer Physics GmbH, various automatic calibration techniques are employed during routine measurements, including noise switching and

calibrations utilizing the internal blackbody (ambient load). These methods are considered relative calibrations, assuming the stability of the noise diode between two absolute calibrations. To uphold absolute accuracy, liquid nitrogen calibrations must be conducted every six months to counteract instrument drifts. Liquid nitrogen (LN2) serves as the cold reference in a hot-cold calibration setup, with the internal load (ambient target) serving as the warm reference temperature. By incorporating a noise diode, four calibration points become available, facilitating determination of all calibration parameters (such as gain, system noise temperature, noise diode temperature, and non-linearity factor) for each channel individually.

MWR Automatic Calibrations

Calibration of MWR profilers is a critical process essential for ensuring accurate and reliable atmospheric measurements. By precisely adjusting the instrument's parameters and responses, calibration enhances the quality and validity of the collected data. This meticulous procedure typically involves various steps, including the characterization of system noise, assessment of instrument stability, and validation against reference standards. Through calibration, uncertainties can be minimized, enabling researchers and meteorologists to confidently interpret and utilize the profiler data for a wide range of applications, including weather forecasting, climate research, and environmental monitoring.

Automatic calibrations for Microwave Radiometer (MWR) systems play a pivotal role in maintaining data accuracy and reliability. These automated procedures are designed to compensate for variations in instrument performance over time, ensuring consistent and precise measurements of atmospheric parameters. Users are well advised to follow the recommendations described in the Operator's manual.

Random error characterization

Characterizing random errors in Microwave Radiometer (MWR) measurements is essential for understanding the uncertainty associated with the data collected. Random errors, also known as stochastic errors, arise from various sources such as electronic noise, atmospheric fluctuations, and instrument imperfections. To characterize these errors, statistical methods such as standard deviation, variance, and correlation analysis are often employed. By analyzing the variability and patterns in the data, researchers can quantify the magnitude of random errors and assess their impact on the overall measurement accuracy. Additionally, techniques such as Monte Carlo simulations may be utilized to model the propagation of random errors throughout data processing and analysis pipelines. Understanding and mitigating random errors in MWR measurements are crucial for improving the reliability and usefulness of the derived atmospheric profiles for applications such as weather forecasting, climate research, and environmental monitoring.

Recommendations for random error characterization: HATPRO: Covariance calculation relative to running mean of HL-temperature. 1hour HL-view with gain calibration every 5 minutes > difference to running mean of HL-temp (20 seconds). variable 14x14x2 (correlation/covariance matrix) > should be performed after every LN2 calibration.

The error covariance matrices provided for the INOE MWR data yield insightful results regarding the random error characterization derived from the 10-minute hotload data. These matrices, computed as 14x14 correlation and covariance matrices, offer a comprehensive depiction of the uncertainties inherent in the measurements. They serve as a crucial tool in understanding the reliability and consistency of the MWR observations, particularly in relation to the calibration process.

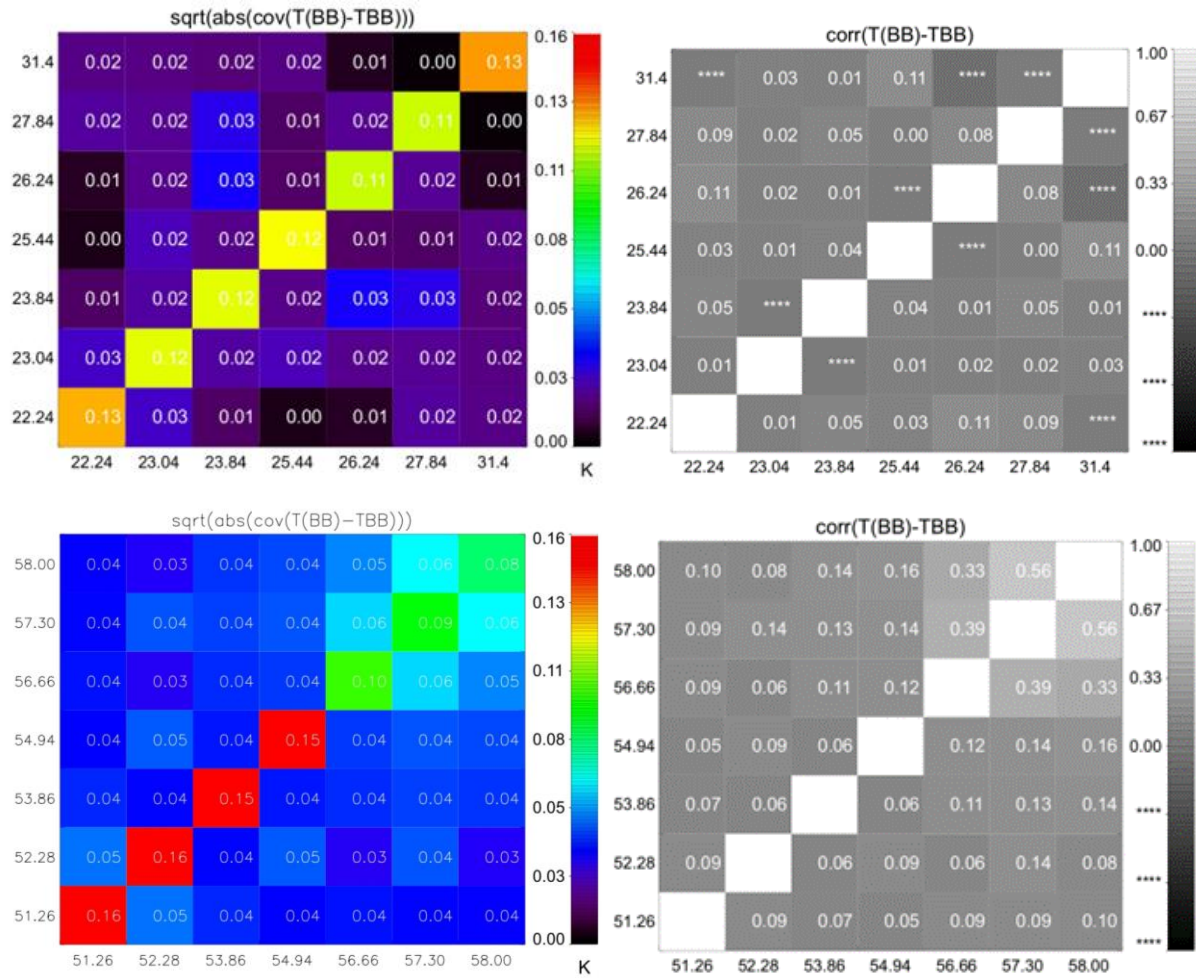


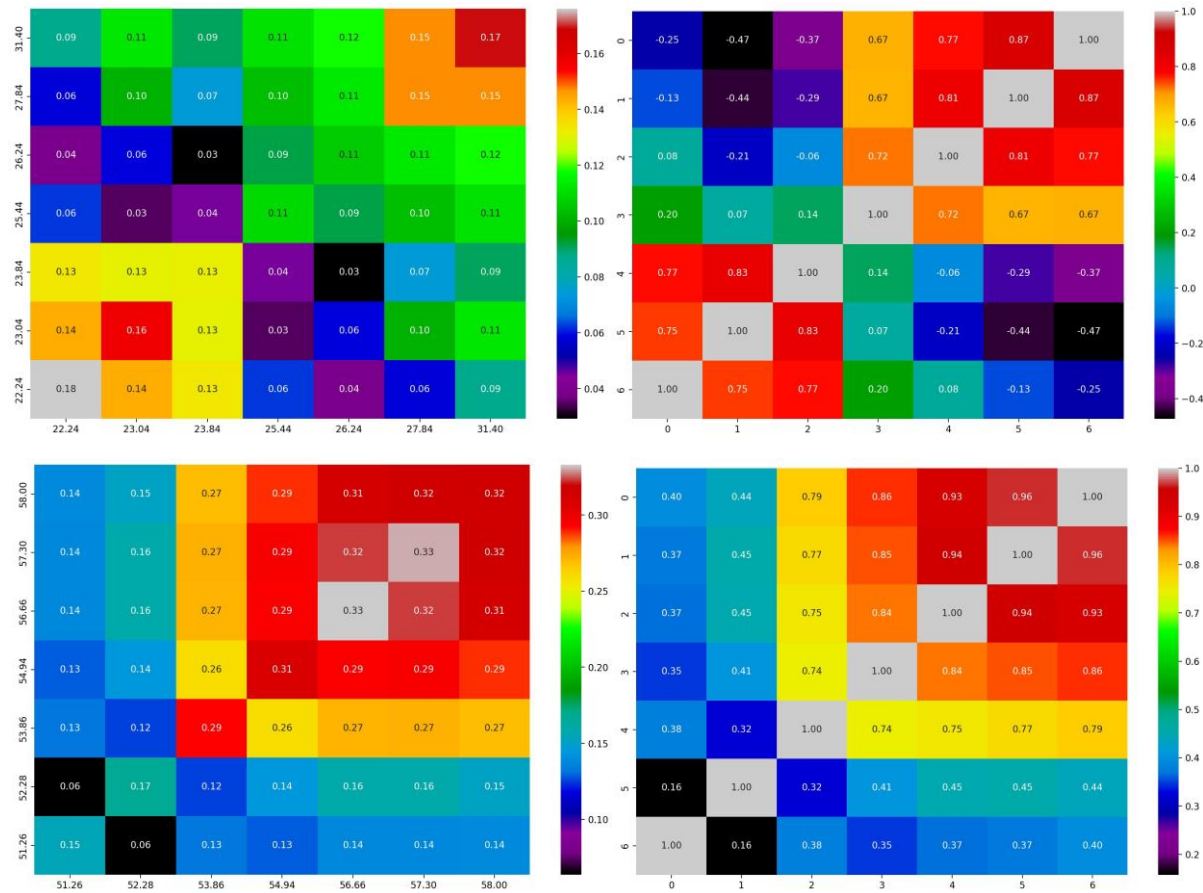
Figure 2. Error covariance matrices INOE MWR – 10 min hotload data

Notably, the independent measurement of the hotload temperature by two internal sensors enhances the robustness of the data. This redundancy ensures a higher level of confidence in the accuracy of the hotload temperature measurements, which are pivotal for calibrating the MWR instrument. The correlation and covariance analysis of the differences between the calculated and measured hotload temperatures provides valuable insights into the effectiveness and reliability of the calibration procedures employed.

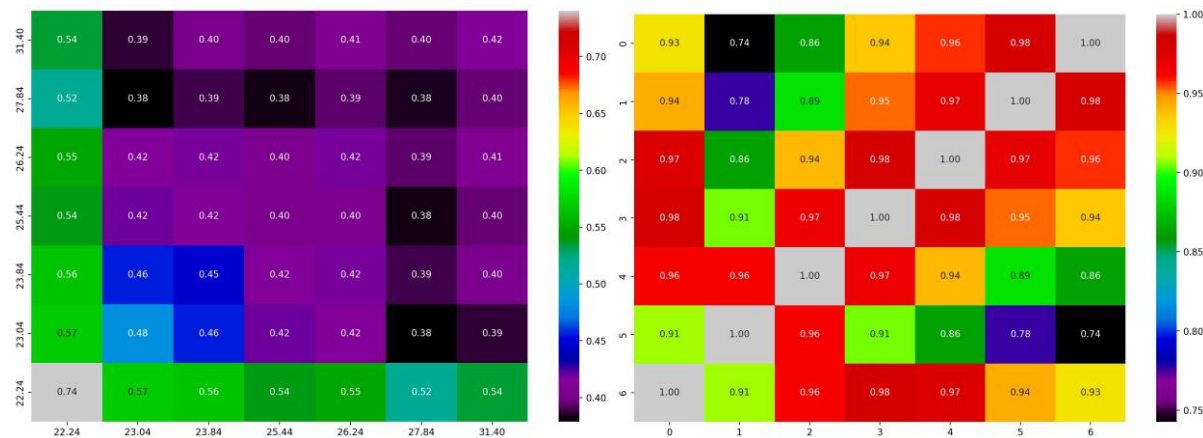
The detailed examination of the 14x14 correlation and covariance matrix reveals significant information about the interrelationships among different error sources within the INOE MWR data. By scrutinizing these matrices, we can discern any discernible patterns or correlations among the measurement errors. These findings are instrumental in refining data processing algorithms, optimizing calibration techniques, and ultimately enhancing the overall quality and reliability of the atmospheric profiles derived from the MWR observations. In summary, the results gleaned from the error covariance matrices offer valuable insights into the random error characterization of the MWR data. They provide essential information for researchers to better understand and interpret the uncertainties associated with the measurements, ultimately contributing to the advancement of atmospheric research and applications.

These matrices encapsulate a wealth of information regarding the statistical properties of INOE MWR measurement errors, offering a comprehensive view of the uncertainties inherent in the datasets. By examining the covariances between different measurement variables, such as brightness temperatures recorded at various frequencies or time intervals, scientists can discern intricate patterns and correlations within the data. This detailed understanding enables the identification of potential sources of error, ranging from instrumental noise and calibration inaccuracies to atmospheric variability and retrieval uncertainties. Consequently, error covariance matrices serve as invaluable diagnostic tools, guiding researchers in optimizing

data collection strategies, refining measurement techniques, and enhancing the overall reliability of MWR observations.

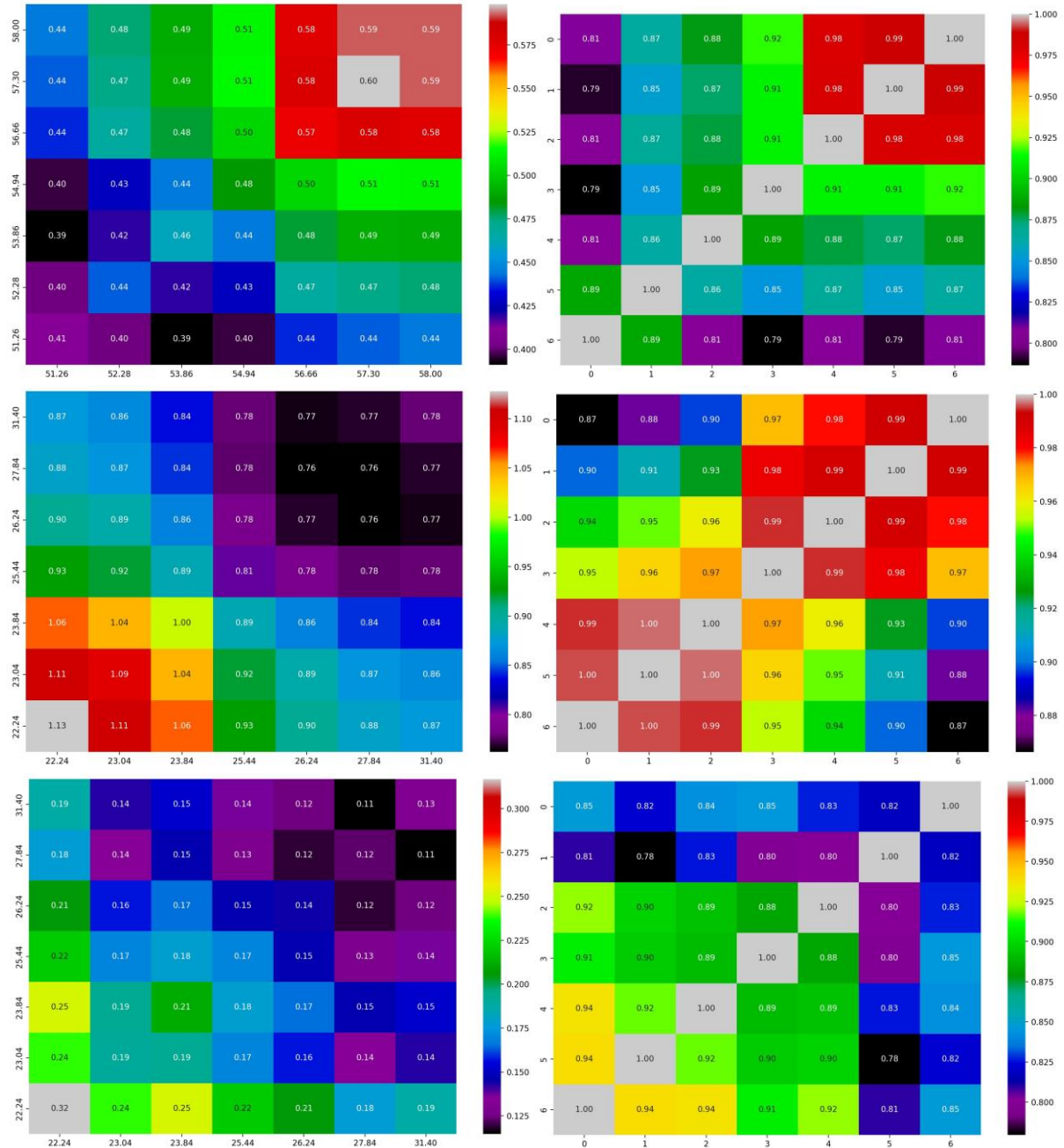


Moreover, error covariance matrices play a pivotal role in the assessment of data quality and the validation of retrieval algorithms used to derive atmospheric parameters from MWR measurements. By quantifying the uncertainties associated with individual observations and their interrelationships, these matrices facilitate the identification of anomalous data points and the detection of systematic biases or trends.



This information is crucial for ensuring the accuracy and consistency of atmospheric profiles generated from MWR data, thereby bolstering the credibility of scientific findings, and supporting informed decision-making in various applications, including weather forecasting, climate research, and environmental monitoring. Additionally, error covariance matrices provide essential inputs for data assimilation techniques, such as Kalman filtering and variational methods, which integrate observational data with numerical models to produce

more accurate and reliable predictions of atmospheric states and processes. Furthermore, the utility of error covariance matrices extends beyond data analysis to the development of innovative data processing algorithms and error estimation methodologies tailored to MWR measurements. By leveraging the rich information encoded in these matrices, researchers can devise sophisticated techniques for mitigating measurement errors, optimizing sensor configurations, and improving the overall robustness of atmospheric retrievals.



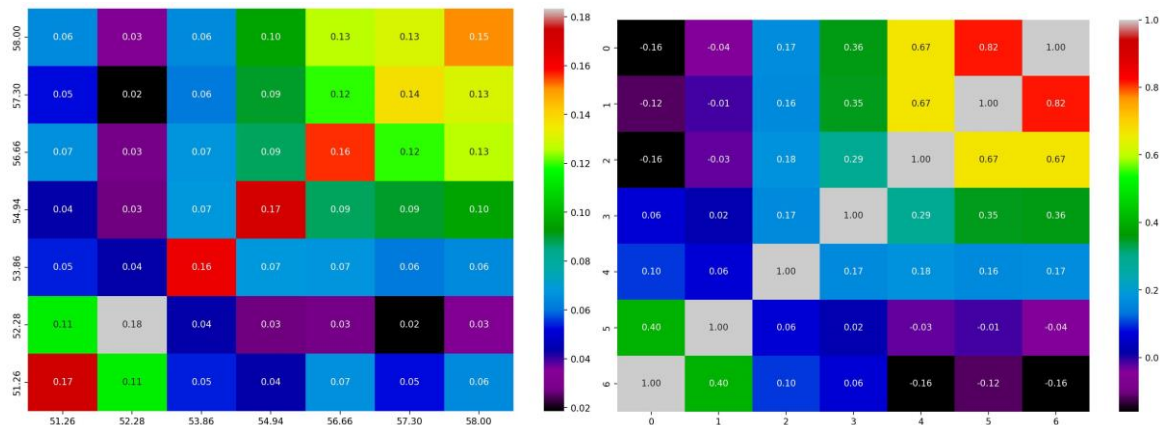


Figure 3. Error covariance matrices for kband and vband INOE MWR – 1,2,3,4 April 2024

For April 1, 2024, the covariance and correlation matrices show that the difference between the measured TB and the calculated TB for channels 1, 2 and 3 in the k-band is almost independent of the difference between the measured T(BB) and the calculated TBB for channels 5, 6 and 7 in the k-band (values close to 0 for covariance and negative or very small values for correlation). For the v-band we have almost independent values for channels 1 and 2. This behaviour can be associated with the presence in the atmosphere of a significant concentration of large aerosol particles (in this case Saharan dust).

For April 3rd and 4th, the correlation matrices show that the difference between T(BB) and TBB for channels 1 and 2 in v-band is anticorrelated with the difference between T(BB) and TBB for channels 3 - 7 in v-band (negative values for correlation). This behaviour could be associated with turbulence in the PBL (PBL instability, precipitation).

This iterative process of refinement and innovation enables continuous advancements in MWR technology and data analysis practices, ultimately enhancing our understanding of the Earth's atmosphere and its complex dynamics. In summary, error covariance matrices serve as indispensable tools for harnessing the full potential of Microwave Radiometer data, driving scientific discovery and innovation in atmospheric science research.

LN2 MWR calibration

Liquid nitrogen (LN2) calibration is a cornerstone of Microwave Radiometer (MWR) instrumentation, playing a pivotal role in ensuring the accuracy and reliability of atmospheric measurements. LN2 serves as a crucial reference standard in calibrating MWR instruments, providing a known and stable cold temperature point against which the radiometer's responses can be calibrated. During LN2 calibration, the MWR's receiver system is subjected to the extreme cold temperature of liquid nitrogen. This process allows for precise calibration of temperature-sensitive components within the instrument.



Figure 4. INOE MWR G2 LN2 calibration - old target



Figure 4. INOE MWR G5 – used radome

By comparing the radiometer's response to the known properties of LN2, calibration parameters such as gain, system noise temperature, and non-linearity factors can be accurately determined. This calibration technique ensures that the MWR instrument maintains consistency and accuracy in its measurements, even in the face of environmental changes and aging effects. Moreover, LN2 calibration enables the MWR to accurately capture atmospheric signals across a wide range of temperatures. This is particularly crucial for applications such as temperature and humidity profiling, where precise measurements are essential for understanding atmospheric processes. By incorporating LN2 calibration into the instrument's routine maintenance schedule, researchers can be confident in the reliability and integrity of the atmospheric data derived from MWR observations.

Recommendations for LN2 calibration (according to TOPROF Recommendations for operation and calibration of Microwave Radiometers (MWR) within a network): Should be performed at least every 6 months and/or after instrument relocation or shutdown; Good weather conditions (low wind speed, rel. humidity < 70%, no precipitation) mandatory! If possible, bring your instrument to a shelter and perform calibration There; Stable operation temperature is needed After moving from outdoor to indoor and before the calibration begins the radiometer needs time for adaption; The bottom of the target must be clean and dry; User should observe the displayed ND values to ensure that they remain stable during the calibration period of about 1 h.

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